DEV CON VIII INDIA

# Networking



## Setting the stage

Yann, then Sukun · Breakout · Part 1 · 2 November 2026 · Room 205A

# Design principles

- **Change little in I***

  DC is already a big consensus change. Networking shouldn’t become the critical path.

- **Minimum viable first**

  Whatever gets us 4× faster finality. Measure it, then tweak it.

- **Easy to scale**

  Fancier things can come later.

# Not in I*.

**We can have it all.**

Just not all at once.



# Here, only the finality gadget

**Available chain**

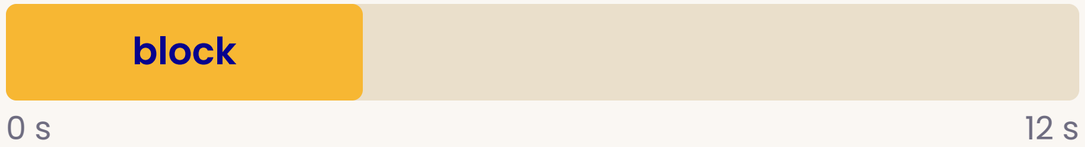
Head votes

**Finality gadget**

Attestations

# Four words

**Slot**




One block. 12 s today.

**Unit**
(subround)




One committee: 4 s of votes, then 4 s of aggregation.

**Round**




Every validator casts one FG vote. 8 slots, 96 s.

**Height**




What the FG votes on. Justified at 2/3 of stake.

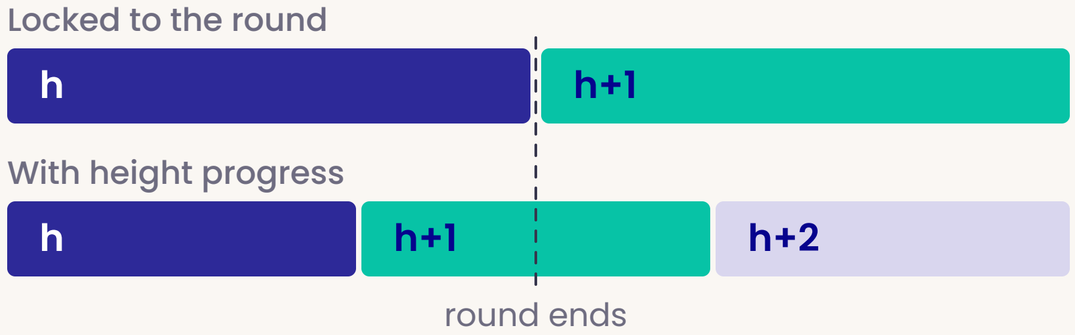
# Two properties we may want




## Decoupling

FG votes keep their own clock, not the slot's.

**In I*: decoupled in the spec, 8 slots in practice**



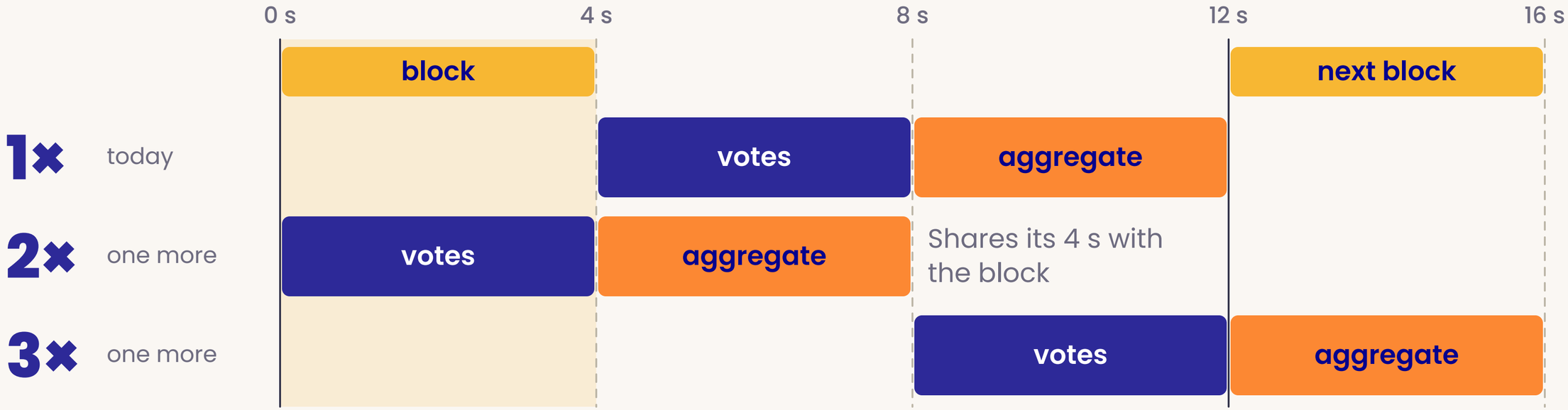

## Height progress

A height can finish mid-round, once 2/3 is in.

**Nice in I*, needed in J***

# Pipelining: 2× to 3×

Today votes use 4 s of every 12 s. Each extra committee adds 4 s.




**We go from 32 subrounds to 23.**

# Bundled attestations: another 2×

EIP-8334: one copy of the data, many signatures.

**4 single attestations · 960 B**




**One bundle of 4 votes · 552 B**

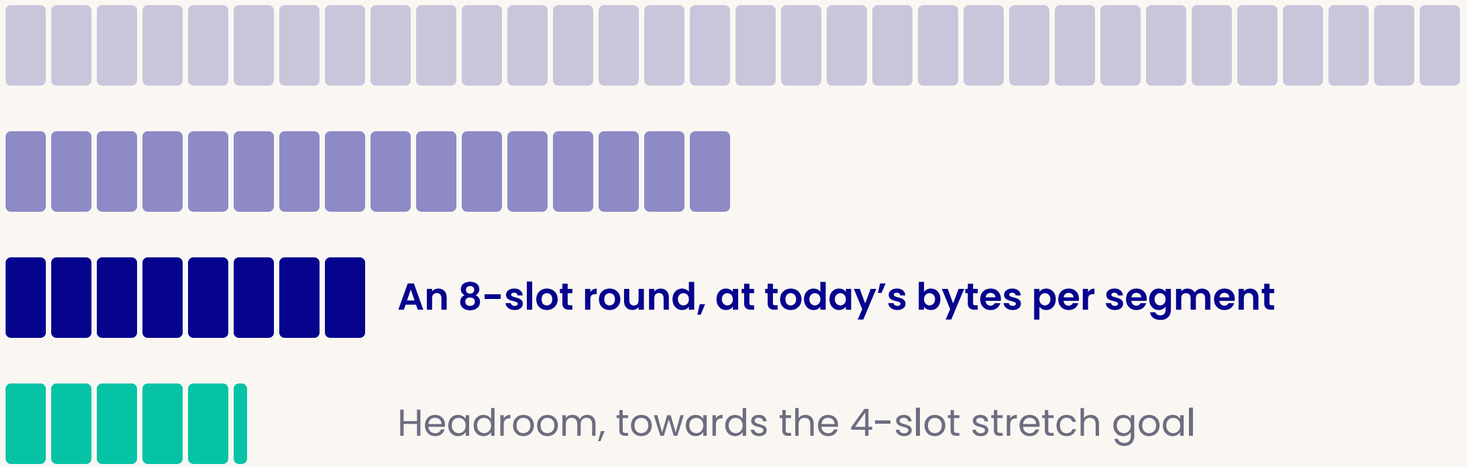



## 240 → 138 B

per vote, in a bundle of 4. Bigger bundles approach 104 B.

# From 32 slots to 8

2× and 2× make the 4× faster finality we're after.




**Fancier things, if we want them later**

Consolidation · Fast path · Large staker prioritization · Committee self-selection

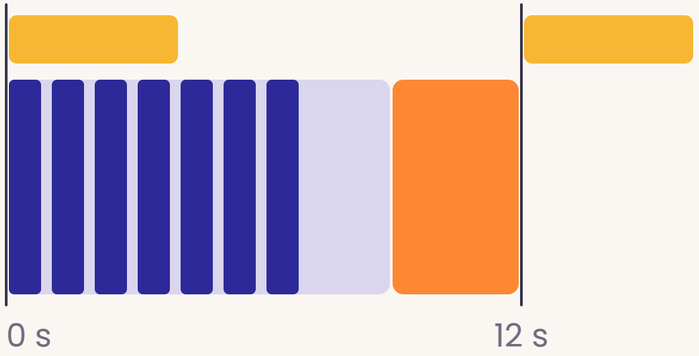
## DC lays the rails.

# Three ways to fill the slot

Not decided yet.

votes aggregation block

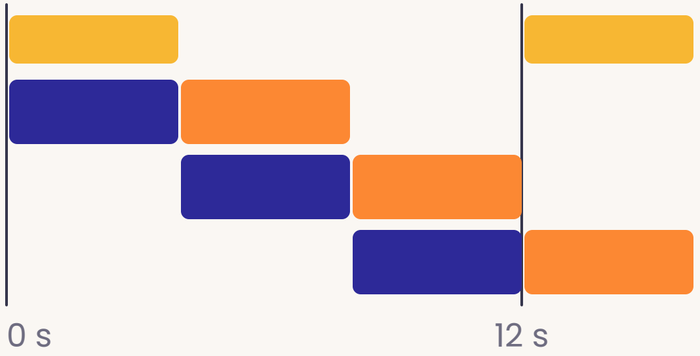
## Staggered publishing




| | |
|---|---|
| Aggregates per slot | **1,024** |
| Vote timing | **honest** |

Simple for one fork. Tied to the slot.

## Multiple committees

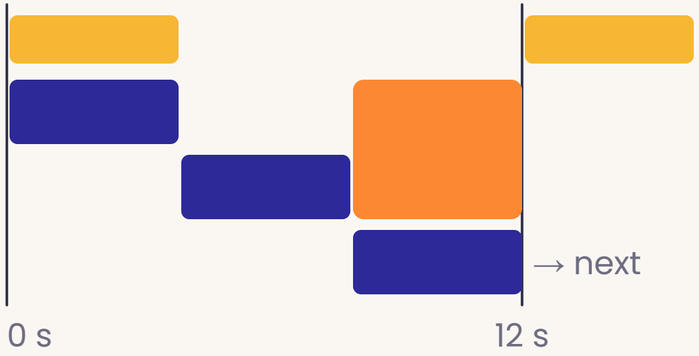



| | |
|---|---|
| Aggregates per slot | **3,072** |
| Vote timing | **enforced** |

Battle-tested shape. 3× the aggregates.

## One aggregation phase

newest

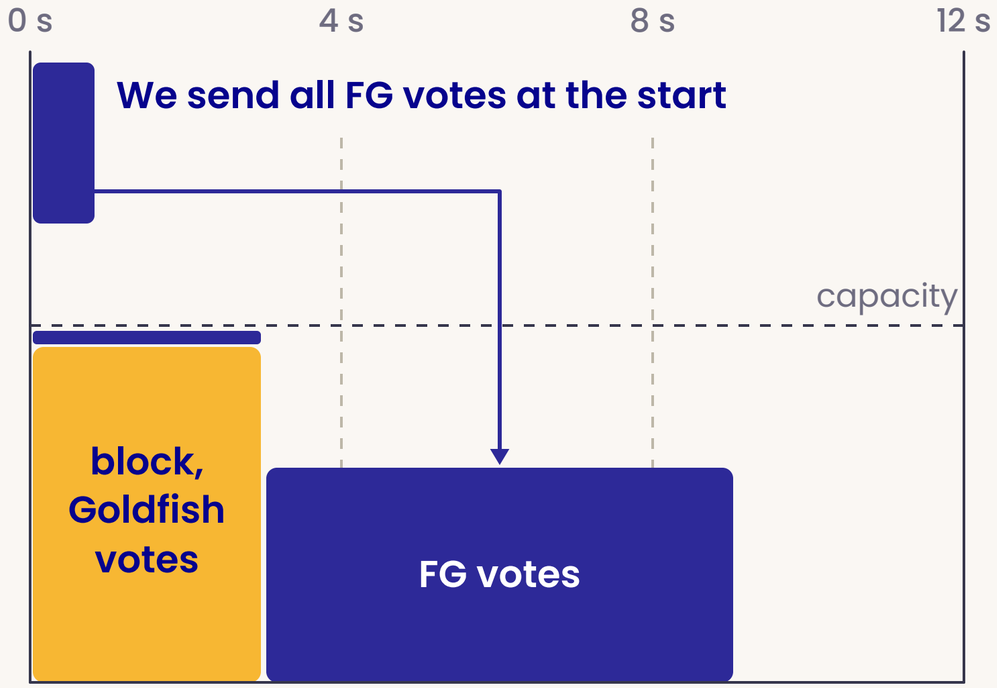



| | |
|---|---|
| Aggregates per slot | **1,024** |
| Vote timing | **enforced** |

Looks like today’s slot. Aggregates span committees.

# Prioritization

Blocks and Goldfish votes first. FG votes fit around them.




**Almost no FG votes alongside Goldfish votes**

## We do it anyway

Independent of the DC spec: no consensus change.

## DC doesn’t rely on it

Minimal spec, on standard networking features.

## Bundling should be enough

Sukun’s numbers are next.

# Properties

From the property doc. Missing one? Let’s add it this week.

## Ground rules for I*

- **Small change**
- **Forward compatible with J***

Marked: the core of the networking layer, from the intro.

## Needed in I* and J*

- Low fork dependence of bitlists
- Fair vote timing
- Fair rewards for validators
- Fair rewards for proposers
- **Uses the whole round**
- **Degrades gracefully**
- Reliable

## Nice in I*, needed in J*

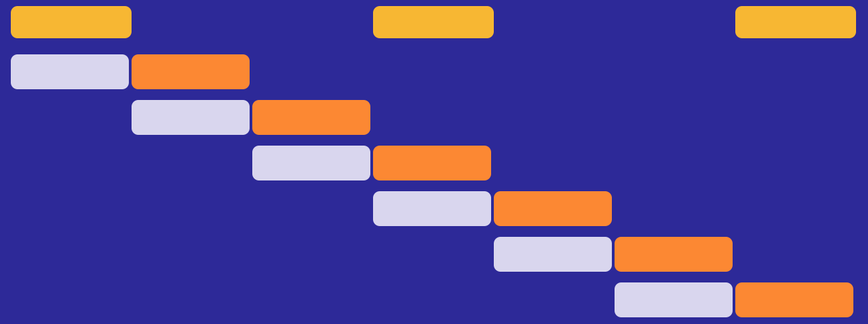
- Fast progress in good conditions
- Enables a safety gadget
- Anonymity enhancing
- Enables aggregation builders
- + Decoupled from the slot?
- + Yours

properties.md, in ethereum/dc-networking

# The setup

- A small change
- About 4×, from two moves
- A few open choices

# Next: Sukun



**Can FG votes be aggregated in time?**

Benchmarks and devnets